

(05-ATI - Pseudocode Writing Task

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① Dijkstra with node penalties:

Problem: Find the shortest path where the total cost includes the edge distance + penalty of each visited node

Pseudocode:

Dijkstra(Graph, source, penalty)

FOR each node v

$\text{dist}[v] \leftarrow \infty$

END FOR

$\text{dist}[\text{source}] \leftarrow 0$

WHILE unvisited nodes exist

u - node with minimum dist

FOR each neighbor v of u

$\text{cost} \leftarrow \text{dist}[u] + \text{weight}(u, v) + \text{penalty}(v)$

IF $\text{cost} < \text{dist}[v]$

$\text{dist}[v] \leftarrow \text{cost}$

END IF

END FOR

END WHILE

RETURN dist

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Output:

Distance[v] $\rightarrow$ minimum total cost to reach node v

parent[v] $\rightarrow$ used to construct the shortest path

② Floyd - Warshall for Intermediate Hub nodes

Problem: Find shortest paths between all pairs of nodes, but allow only a specific subset of Hub nodes to be used as intermediate nodes.

Pseudocode:

FloydWarshall (Graph, Hubs)

Initialize dist[i][j] with edge weights

FOR each hub k in hubs

FOR each node i

FOR each node j

$\text{dist}[i][j] \leftarrow \min[\text{dist}[i][j], \text{dist}[i][k] + \text{dist}[k][j]]$

END FOR

END FOR

END FOR

RETURN dist

Key idea:

Unlike normal Floyd - Warshall, which uses every node as a possible intermediate node. This algorithm uses only the nodes in hubnodes.

Output:

Distance $[i][j]$ gives the shortest path cost from node i to node j , using only the Specified hub nodes as intermediate nodes